1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 34 · Transformations of Functions

Suppose $f(x) = x^2$



Sketch $y = f(x)$:

Sketch $y = f(x + 2)$

What do you notice about the relationship between the graphs of $y = f(x)$ and $y = f(x + 2)$?

Chapter 4 · Original slide 35 · Transformations of Functions

This is all you need to remember when considering how transforming your function transforms your graph...

Affects which axis?

What we expect or opposite?

Change inside $f()$

Change outside $f()$

Therefore...

$y = f(x - 3)$

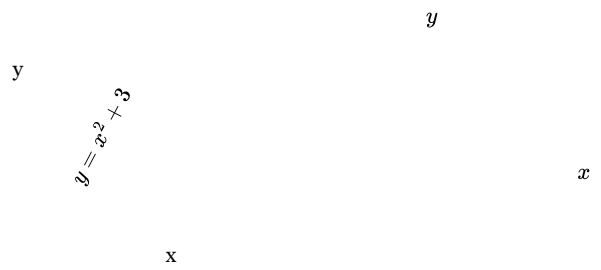
$y = f(x) + 4$

$y = f(5x)$

$y = 2f(x)$

1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 36 · Sketching transformed graphs



Sketch $y = \frac{2}{x+1}$

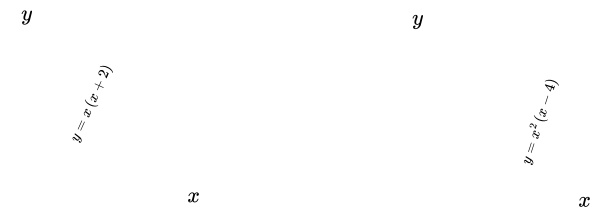
Sketch $y = x^2 + 3$

Imagine a sketch of $y = x^2$ and then do the translation, ensuring you adjust any intercepts with the axes.

The transformation might result in new intercepts or roots. You can find these in the usual way. Do not forget them!

Draw asymptotes using a dotted line and write its equation on it.

Chapter 4 · Original slide 37 · More Examples



Sketch $y = x(x+2)$. On the same axes, sketch $y = (x-a)(x-a+2)$, where $a > 2$.

Sketch $y = x^2(x-4)$. On the same axes, sketch the graph with equation

$y = (2x)^2(2x-4)$

$y = (x-a)(x-a+2)$

$y = (2x)^2(2x-4)$

Note that our intercepts are in terms of a.

1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 38 · Reflections of Graphs

If $y = x(x+2)$, sketch $y = f(x)$ and $y = -f(x)$ on the same axes.

You did this at GCSE. The minus is outside the function, so affects the output, i.e. the y value.

The y values are negated, resulting in a reflection in the x-axis.

y

$y = f(x)$

$\mathbf{y= -f(x)}$

x

Chapter 4 · Original slide 39 · Test Your Understanding



If $y = (x+1)(x-2)$, sketch $y = f(x)$ and $y = f(\frac{x}{3})$ on the same axes.

Sketch the graph of $y = \frac{2}{x} + 1$, ensuring you indicate any intercepts with the axes.

1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 40 · Exercise 4E – Pure Y1 page 74-75 (translations)

7 The point P(4,-1) lies on the curve with equation $y = \mathrm{f}(x)$.

a State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x - 2)$.
b State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(x) + 3$.

8 The graph of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = \frac{1}{x}$ is translated so that the asymptotes are at $x = 4$ and $y = 0$. Write down the equation for the transformed function in the form $y = \frac{1}{x+a}$.

Chapter 4 · Original slide 41 · Exercise 4E – Pure Y1 page 74-75 (translations)

11 a Sketch the graph of $y = x^3 + 4x^2 + 4x$.
b The point with coordinates (-1,0) lies on the curve with equation $y = (x + a)^3 + 4(x + a)^2 + 4(x + a)$ where a is a constant. Find the two possible values of a.

1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 42 · Exercise 4E – Pure Y1 page 74-75 (translations)

12 a Sketch the graph of $y = x(x+1)(x+3)^2$.
b Find the possible values of b such that the point (2,0) lies on the curve with equation $y = (x+b)(x+b+1)(x+b+3)^2$.

Chapter 4 · Original slide 43 · Exercise 4F – Pure 1 Page 78 (stretches/reflections)

7 The point P(2,-3) lies on the curve with equation $y = \mathrm{f}(x)$.

a State the coordinates that point P is transformed to on the curve with equation $y = \mathrm{f}(2x)$.
b State the coordinates that point P is transformed to on the curve with equation $y = 4\mathrm{f}(x)$.

8 The point Q(-2,8) lies on the curve with equation $y = \mathrm{f}(x)$. State the coordinates that point Q is transformed to on the curve with equation $y = \mathrm{f}(\frac{1}{2}x)$.

1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 44 · Exercise 4F – Pure 1 Page 78 (stretches/reflections)

9 a Sketch the graph of $y = (x-2)(x-3)^2$.
b The graph of $y = (ax-2)(ax-3)^2$ passes through the point (1,0). Find two possible values for a.

Chapter 4 · Original slide 45 · Effect of transformation on specific points

Sometimes you will not be given the original function, but will be given a sketch with specific points and features you need to transform.

Where would each of these points end up?

$\mathbf{y=}f(x)$	$(\mathbf{4},\mathbf{3})$	$(\mathbf{1},\mathbf{0})$	$(\mathbf{6},\mathbf{-4})$
$y = f(x+1)$			
$y = f(2x)$			
$y = 3f(x)$			
$y = f(x) - 1$			
$y = f\left(\frac{x}{4}\right)$			
$y = f(-x)$			
$y = -f(x)$			

1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 46 · Test Your Understanding

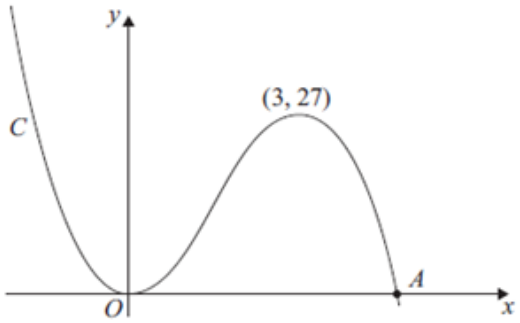


Figure 1

Figure 1 shows a sketch of the curve C with equation $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = x^2(9 - 2x)$. There is a minimum at the origin, a maximum at the point (3,27) and C cuts the x-axis at the point A.

(a) Write down the coordinates of the point A. (1)
(b) On separate diagrams sketch the curve with equation
(i) $y = \mathrm{f}(x + 3)$,
(ii) $y = \mathrm{f}(3x)$.
On each sketch you should indicate clearly the coordinates of the maximum point and any points where the curves cross or meet the coordinate axes. (6)
The curve with equation $y = \mathrm{f}(x) + k$, where k is a constant, has a maximum point at $(3, 10)$.
(c) Write down the value of k. (1)

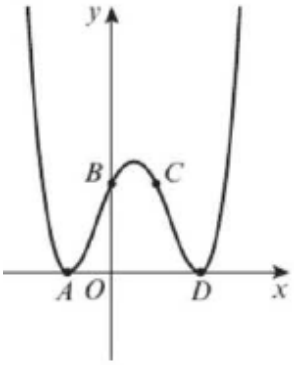
Chapter 4 · Original slide 47 · Exercise 4G Pure Y1 Pages 80

5 The point P(2,1) lies on the graph with equation $y = \mathrm{f}(x)$.

a On the graph of $y = \mathrm{f}(ax)$, the point P is mapped to the point $Q(4, 1)$. Determine the value of a.
b Write down the coordinates of the point to which P maps under each transformation
i $\mathrm{f}(x - 4)$ **ii** $3\mathrm{f}(x)$ **iii** $\frac{1}{2}\mathrm{f}(x) - 4$

1.28 Graph Transformations: Single Transformations

Chapter 4 · Original slide 48 · Exercise 4G Pure Y1 Pages 80



6 The diagram shows a sketch of a curve with equation $y = \mathrm{f}(x)$. The points A(-1,0),B(0,2),C(1,2),D(2,0) lie on the curve. Sketch the following graphs and give the coordinates of the points A,B,C,D after each transformation:

a $y + 2 = \mathrm{f}(x)$ **b** $\frac{1}{2}y = \mathrm{f}(x)$ **c** $y - 3 = \mathrm{f}(x)$ **d** $3y = \mathrm{f}(x)$

e $2y - 1 = \mathrm{f}(x)$